

Automated IoT Smart Water Meter

Project Report



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1 Problem Statement

Sri Lanka's water supply network continues to experience significant Non-Revenue Water (NRW) losses, resulting in substantial financial losses, operational inefficiencies, and reduced water availability for consumers. NRW refers to treated water that is produced but not billed due to physical leaks, unauthorized consumption, inaccurate metering, or administrative errors. According to the National Water Supply and Drainage Board (NWSDB), the national NRW level remains approximately 24.8, while the Auditor General's report estimates losses of nearly 27. It is estimated that every 1% reduction in NRW could save approximately Rs. 790 million annually, demonstrating the significant economic impact of improving water management.

Traditional mechanical water metering systems rely heavily on manual meter reading, making the billing process susceptible to human error, estimated readings, delayed data collection, and meter tampering. Furthermore, consumers and water authorities lack real-time visibility into water consumption, preventing the early identification of abnormal usage patterns and leakage.

Another major limitation of the current system is the absence of remote monitoring and control capabilities. Water utilities must physically dispatch maintenance crews to inspect meters, investigate leaks, or disconnect water supplies for maintenance or unpaid bills. This process is time-consuming, labour-intensive, and increases operational costs.

Maintenance expenses associated with aging mechanical water meters also place a considerable financial burden on utilities. A JICA pilot study conducted in the Harispattuwa Division reported an 8.37% defective meter rate and approximately Rs. 7.9 million spent on repairs, illustrating the high maintenance requirements of conventional metering infrastructure.

These challenges highlight the need for a modern, intelligent, and cost-effective smart water metering solution capable of providing real-time monitoring, automated billing, leak detection, remote communication, and cloud-based data management. Such a system would reduce NRW, improve billing accuracy, lower operational costs, and enhance customer service while supporting the digital transformation of Sri Lanka's water distribution network.

2 Objectives

The primary objective of this project is to design and develop a low-cost, IoT-enabled smart water metering system capable of improving the efficiency, accuracy, and reliability of water distribution and billing. The specific objectives are:

- Design and develop a low-cost, battery-powered IoT smart water meter suitable for residential water distribution systems providing an expected operational lifetime of more than 10 years under normal operating conditions.
- Enable real-time remote transmission of water flow rate and cumulative consumption data to a centralized cloud database using GSM communication.
- Automate the water billing process using accurate, tamper-resistant consumption measurements, thereby eliminating manual meter reading.
- Develop web-based dashboards for both utility administrators and consumers to monitor live water consumption, billing information, revenue statistics, and network-wide analytics.
- Support the reduction of NRW through continuous flow monitoring, leak detection, and anomaly identification based on real-time consumption patterns.

3 Survey & Market Validation

To validate the problem and gauge demand for a smart metering solution, the team conducted a survey targeting household water users. The results reinforce the core assumptions behind this proposal: water wastage is common, manual management remains the norm, and there is strong demand for real-time, remote monitoring.

3.1 Current Water Supply Management

Q1 — How do you currently manage your water supply?

A significant proportion of respondents still rely on manual methods for managing their water supply. Around 45.5% use a manual valve, while 31.8% operate a manually-switched electric pump. Together, over three-quarters of respondents currently have no automated or remote control over their household water supply.

1. How do you currently manage water supply?

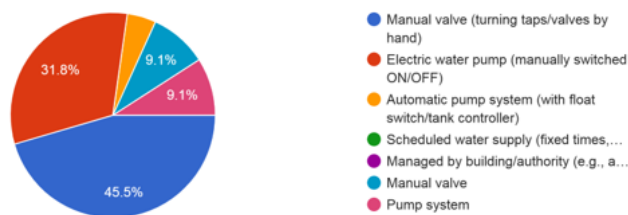


Figure 1: Current methods used by households to manage water supply

3.2 Household Water Wastage Issues

Q2 — Do you face water wastage issues?

The survey revealed that water wastage is a widespread issue, with 72.7% of respondents reporting that they experience water wastage problems in their households.

2. Do you face water wastage issues?

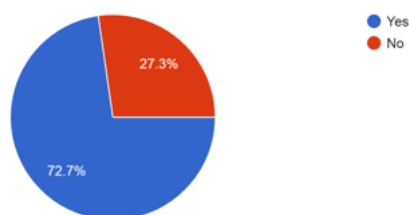


Figure 2: Percentage of households experiencing water wastage issues

Q3 — If yes, what kind of issues?

Among respondents who experienced water wastage, forgetting to turn off water was identified as the primary cause, accounting for 47.1% of responses. Leakage was the second most common issue (23.5%), followed by overuse (17.6%). These issues can be directly addressed through the proposed system using remote shutoff capability and automated leakage alerts.

3. If yes, what kind of issues?

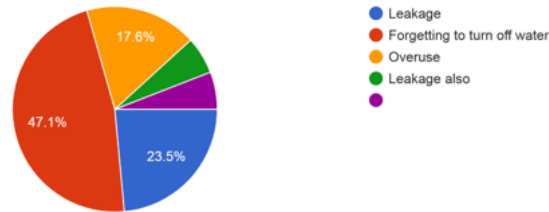


Figure 3: Main causes of household water wastage

3.3 Demand for Real-Time Monitoring**Q5 — Would you like to monitor water usage in real time?**

The survey demonstrated strong demand for real-time monitoring, with 90.9% of respondents expressing interest in monitoring their household water consumption. This directly validates the core value proposition of the AquaSync consumer application.

5. Would you like to monitor water usage in real time?

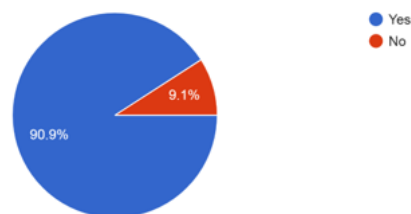


Figure 4: User interest in real-time water usage monitoring

3.4 Perceived Usefulness of the System**Q6 — How useful would this system be for you?**

The proposed smart water metering system received highly positive feedback. Approximately 95% of respondents rated the system as either “very useful” (approximately 50%) or “useful” (approximately 45%), indicating strong user acceptance.

6. How useful would this system be for you?

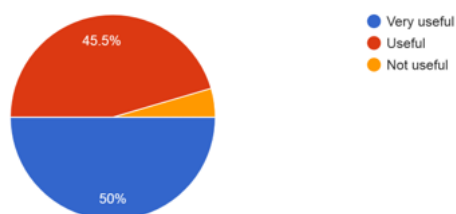


Figure 5: User perception of the usefulness of the proposed system

3.5 Preferred Features

Q7 — Which features do you prefer?

Leak detection alerts were the most requested feature, selected by 68.2% of respondents. Monthly usage reports and real-time water usage monitoring were preferred by 63.6% and 59.1% of respondents, respectively. Furthermore, remote ON/OFF valve control and automatic shutoff features were valued by 54.5% of respondents each.

7. Which features do you prefer?

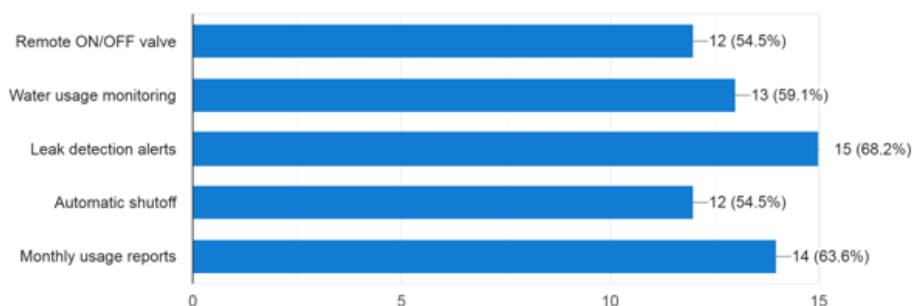


Figure 6: Preferred features of the smart water metering system

3.6 Customer Willingness to Pay

Q8 — Would you be willing to pay for such a system? (22 responses)

Approximately 73% of respondents indicated that they would be willing to pay for the proposed smart water metering system. This result provides evidence of commercial viability and suggests that the solution has potential beyond a purely utility-driven deployment.

8. Would you be willing to pay for such a system?

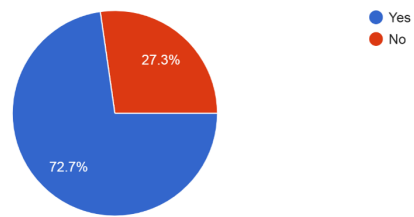


Figure 7: Customer willingness to pay for the proposed system

4 Stakeholder and Needs Analysis

The proposed Automated IoT Smart Water Meter is intended to benefit multiple stakeholders by addressing their specific operational and service requirements.

Table 1: Stakeholder and Needs Analysis

Stakeholder	Key Needs
NWSDB / Utility	• Real-time monitoring of NRW • Automated leak detection and anomaly identification • Accurate and tamper-resistant billing data • Reduced operation and maintenance (O&M) costs
Consumers / Households	• Transparent visibility of water consumption • Accurate and dispute-free billing • Mobile access to real-time consumption data • Notifications for remote disconnection and abnormal usage
Government / Regulators	• Achievement of national NRW reduction targets • Availability of auditable metering and billing records • Reduced public subsidy burden • Alignment with Sustainable Development Goal (SDG) 6 for clean water and sanitation

5 Proposed Solution and Core Mechanism

The proposed Automated IoT Smart Water Meter provides an end-to-end solution for intelligent water consumption monitoring, automated billing, and NRW reduction. The system integrates flow sensing, embedded processing, wireless communication, cloud storage, and a web-based dashboard into a single IoT platform. The overall operation of the system is summarized below.

- **Flow Sensing:** A water flow sensor containing a rotor with an embedded permanent magnet generates electrical pulses as water passes through the pipe. A Hall-effect sensor detects these pulses, where the pulse frequency is directly proportional to the instantaneous water flow rate.
- **Microcontroller Processing:** An STM32L0 ultra-low-power microcontroller counts the incoming pulses and computes the instantaneous flow rate and cumulative water volume using the sensor calibration factor.
- **GSM Communication:** A SIM800L GSM module periodically transmits the measured data to the cloud using HTTP POST requests over the cellular network, enabling remote monitoring without requiring local Wi-Fi infrastructure.
- **Cloud Backend:** A Supabase cloud database stores each meter reading together with the corresponding meter identification number and timestamp, ensuring secure data storage, traceability, and auditability.
- **Billing and User Dashboard:** A web-based consumer dashboard retrieves the stored data to provide real-time consumption visualization, estimated billing, historical usage analysis, and leak notifications for both consumers and utility administrators.

5.1 Key Features

The proposed system offers several features that improve the efficiency, reliability, and usability of conventional water metering systems.

- **Real-Time Monitoring:** Live flow rate and cumulative consumption data are transmitted to the cloud via GSM, eliminating the need for manual meter reading.
- **Automated Billing:** Water bills are generated automatically using accurate and tamper-resistant consumption records stored in the cloud database.

- **Battery-Powered Operation:** The device is powered by high-energy-density Li-SOCl_2 primary batteries, enabling long-term standalone operation without connection to the electrical grid.
- **NRW Detection:** Continuous flow monitoring allows the detection of abnormal consumption patterns and potential leaks, helping utilities reduce NRW.
- **Cloud-Based Data Management:** All meter readings are securely stored in a centralized cloud database, enabling historical analysis, auditing, and remote access.
- **Consumer Dashboard:** A user-friendly web application allows consumers to monitor water usage, track estimated bills, view historical consumption trends, and receive notifications of abnormal water usage.

6 Hardware Architecture

The proposed smart water meter is built using low-power embedded hardware components selected to maximize battery life, reliability, and communication capability while minimizing deployment and maintenance costs.

Table 2: Hardware Architecture

Component	Role	Specification
STM32L0 MCU	Microcontroller	32-bit ARM Cortex-M0+ running at 32 MHz, 64 KB Flash, 8 KB SRAM, I ² C, SPI, UART, LPUART and LPTIM peripherals, compatible with STM32CubeIDE and Arduino framework.
SIM800L GSM Module	Communication	Quad-band GSM/GPRS communication module supporting AT commands, HTTP POST requests to the cloud backend, and ultra-low-power sleep mode.
Li-SOCl ₂ Battery	Power Supply	3.6 V primary (non-rechargeable) lithium-thionyl chloride battery with ultra-low self-discharge (<1% per year), optimized for low-power duty-cycled operation and an expected operational lifespan exceeding 10 years.
Hall-Effect Flow Sensor	Flow Sensing	Rotor-mounted permanent magnet and Hall-effect sensor producing pulses proportional to water flow, enabling accurate measurement of instantaneous flow rate and cumulative water volume without additional sensing hardware.

6.1 Alternatives Considered

Several alternative hardware solutions were evaluated during the design process before selecting the final system architecture.

- **Li-ion Battery (3.6–4.2 V, 1500–3600 mAh):** Rejected in favor of the Li-SOCl₂ battery due to its significantly longer service life, lower self-discharge rate, and maintenance-free operation suitable for sealed outdoor installations.
- **ATmega328P Microcontroller (16 MHz, 32 KB Flash, 2 KB SRAM):** Rejected because the STM32L0 provides substantially lower power consumption, greater processing capability, larger memory, and a richer set of peripherals while operating at 3.3 V.

- **Solar Power Supply (5 V, 200 mA):** Considered as an alternative power source but rejected because underground or enclosed water meter installations often lack reliable sunlight, reducing system availability and increasing installation complexity.

6.2 Power Consumption and Battery Life Estimation

To validate the system's capacity for long-term standalone operation, a theoretical battery life estimation was conducted. The microcontroller alternates between an ultra-low-power sleep state (waking only for RTC and sensor interrupts) and an active transmission state where the SIM800L GSM module connects to the network.

The time-weighted average current consumption, I_{avg} , over a standard 1-hour (3,600 s) reporting cycle is calculated using the following formula:

$$I_{\text{avg}} = \frac{(I_{\text{tx}} \times t_{\text{tx}}) + (I_{\text{sleep}} \times t_{\text{sleep}})}{T_{\text{cycle}}} \quad (1)$$

Based on component specifications and transmission timing:

- **Transmission State:** $I_{\text{tx}} = 250 \text{ mA}$ for a duration of $t_{\text{tx}} = 10 \text{ s}$
- **Sleep State:** $I_{\text{sleep}} = 1.5 \text{ } \mu\text{A} = 0.0015 \text{ mA}$
- **Sleep Time:** $t_{\text{sleep}} = 3,600 - 10 = 3,590 \text{ s}$

Substituting these values yields:

$$\begin{aligned} I_{\text{avg}} &= \frac{(250 \times 10) + (0.0015 \times 3590)}{3600} \\ &= \frac{2500 + 5.385}{3600} \\ &\approx 0.696 \text{ mA} \end{aligned} \quad (2)$$

The system is powered by a 19,000 mAh Li-SOCl₂ primary battery. Assuming a conservative 85% usable capacity (16,150 mAh) to account for voltage drops and temperature fluctuations, the expected lifespan under an hourly GSM transmission cycle is:

$$\text{Lifespan} = \frac{16,150 \text{ mAh}}{0.696 \text{ mA} \times 24 \text{ h/day} \times 365 \text{ days/year}} \approx 2.65 \text{ years} \quad (3)$$

Note: While hourly transmission yields a lifespan of approximately 2.65 years, the operational objective of 10+ years can be achieved by adjusting the firmware duty cycle to transmit

data daily rather than hourly, drastically reducing the GSM module's active time.

7 PCB Design and Schematic Architecture

7.1 Top-level schematic hierarchy

The PCB was designed using Altium Designer with a hierarchical architecture to improve modularity, readability, and ease of verification. The top-level schematic references four functional sub-sheets: **Power**, **Microcontroller (MCU)**, **Communication**, and **Peripherals**. These sheets are interconnected using labeled ports such as GSM_ON/OFF, USART1, TIM2 channels, and I2C1, allowing each subsystem to be developed, tested, and verified independently while maintaining a clean overall design.

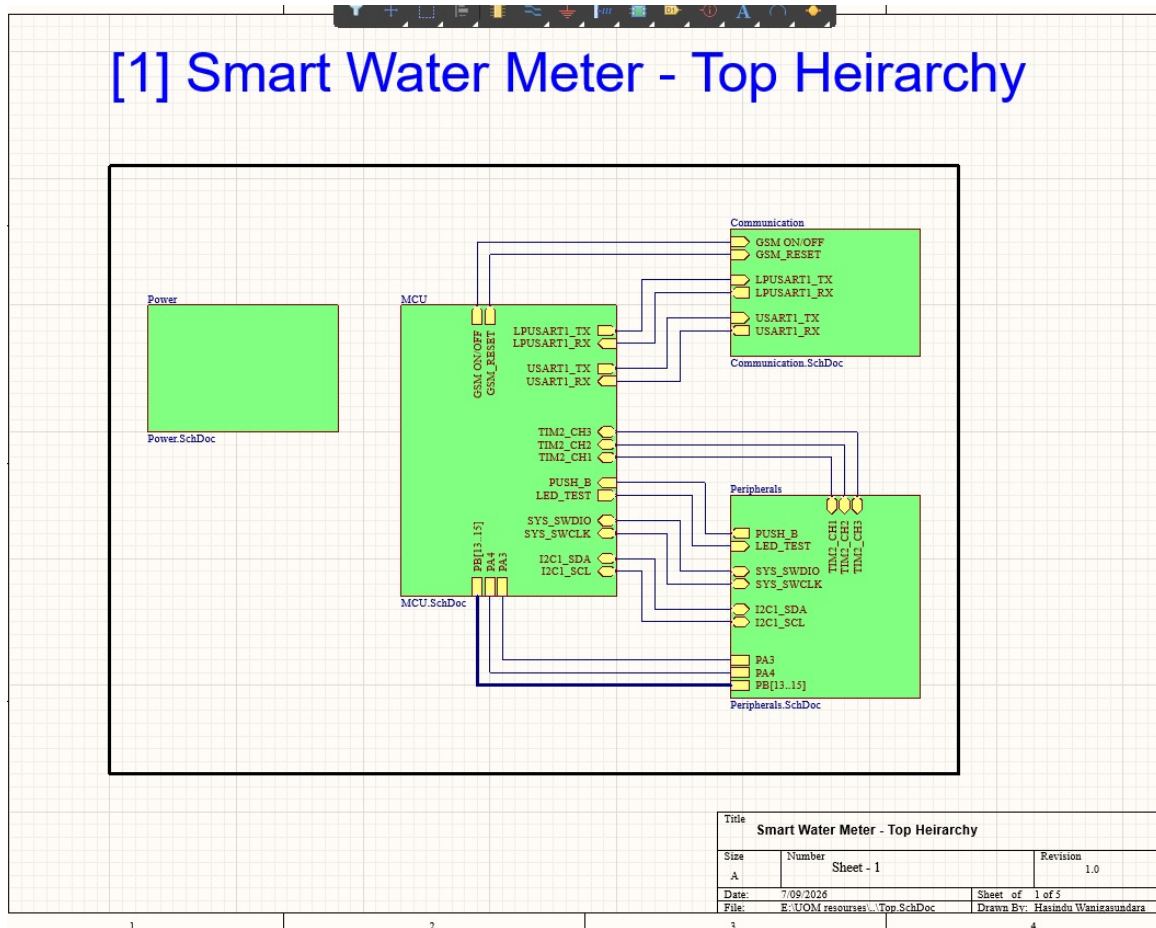


Figure 8: Top-level schematic hierarchy showing the Power, MCU, Communication, and Peripherals sheets.

7.2 Power Subsystem

The power subsystem generates two regulated supply rails from the battery source to satisfy the different voltage requirements of the system components.

- **3.3 V LDO Regulator (XC6206P332MR):** Supplies the STM32L0 microcontroller and other low-power peripherals from the battery voltage. The regulator is stabilized using 22 μF input and output filter capacitors.
- **4 V Buck–Boost Converter (TPS61023DRLR):** Generates a stable 4 V supply for the SIM800L GSM module using a 2.2 μH inductor and a resistive feedback network to set the output voltage.

The output voltage of the TPS61023 buck–boost converter is determined by the external feedback resistor divider according to

$$V_{\text{OUT}} = V_{\text{REF}} \left(1 + \frac{R_{\text{TOP}}}{R_{\text{BOTTOM}}} \right)$$

where the internal reference voltage is $V_{\text{REF}} = 0.6 \text{ V}$.

Using

$$R_{\text{TOP}} = 68 \text{ k}\Omega, \quad R_{\text{BOTTOM}} = 12 \text{ k}\Omega,$$

the output voltage is

$$V_{\text{OUT}} = 0.6 \left(1 + \frac{68}{12} \right) = 0.6 \left(\frac{80}{12} \right) = 4.0 \text{ V}.$$

Therefore, the feedback resistor network is designed to produce a regulated output voltage of **4.0 V**, which supplies the SIM800L GSM module.

- **Battery Connector (J20):** Provides the battery interface and includes a status LED with a current-limiting resistor together with a Schottky diode for reverse-polarity protection.

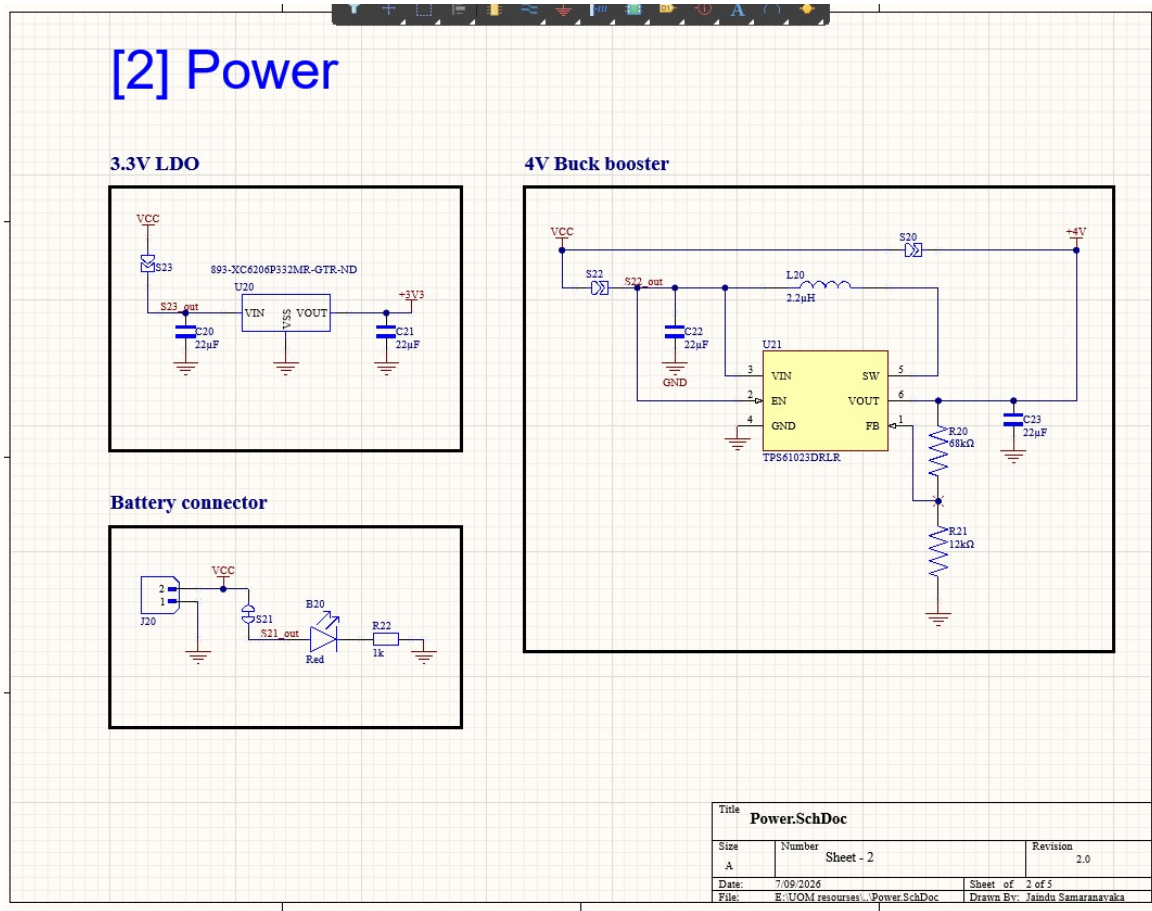


Figure 9: Power subsystem showing the 3.3 V LDO regulator, 4 V buck–boost converter, and battery interface.

7.3 Microcontroller (MCU)

The STM32L071CBT6, based on the ARM Cortex-M0+ architecture, serves as the central controller of the smart water meter. It interfaces with all system modules through dedicated communication and peripheral interfaces.

Major interfaces include:

- TIM2 input capture channels for Hall-effect sensor pulse counting.
- USART1 and LPUART1 for GSM communication.
- I²C1 interface for the OLED display.
- Serial Wire Debug (SWD) interface for firmware programming and debugging.
- General-purpose I/O for the push button, status LED, and GSM power-control signals.

To ensure stable operation during low-power modes, the MCU is supported by multiple decoupling capacitors (C33–C38) and a 32.768 kHz crystal oscillator (OSC30) with load capacitors C30 and C31 for the real-time clock (RTC).

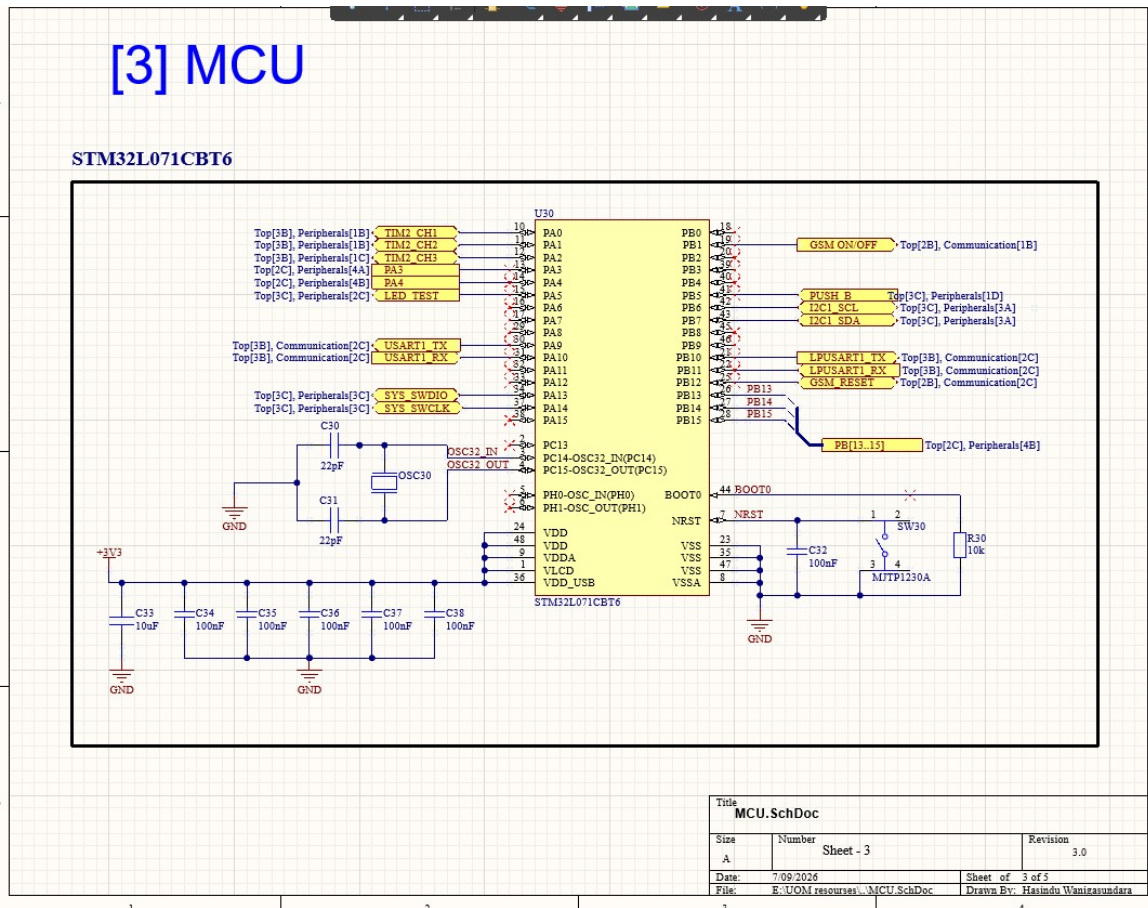


Figure 10: Microcontroller subsystem featuring the STM32L071CBT6 with SWD, USART, TIM2, and I²C interfaces.

7.4 Communication Subsystem

Wireless communication is provided by the SIM800L quad-band GSM module, enabling periodic transmission of water consumption data to the cloud.

To minimize energy consumption, the GSM module is powered only during transmission cycles. An AO3401A P-channel MOSFET (Q40), controlled through an MMBT3904 NPN transistor (Q41), allows the STM32L0 to completely disconnect power from the GSM module when it is idle.

A 4700 μ F bulk capacitor (C40) supplies the high peak currents required during GSM transmission bursts, while Schottky diodes on the RX, TX, and RESET lines protect the MCU's 3.3 V logic from higher voltage levels generated by the communication module.

7.5 Peripheral Subsystem

The peripheral subsystem integrates all external interfaces required for sensing, user interaction, programming, and future expansion.

[4] Communication

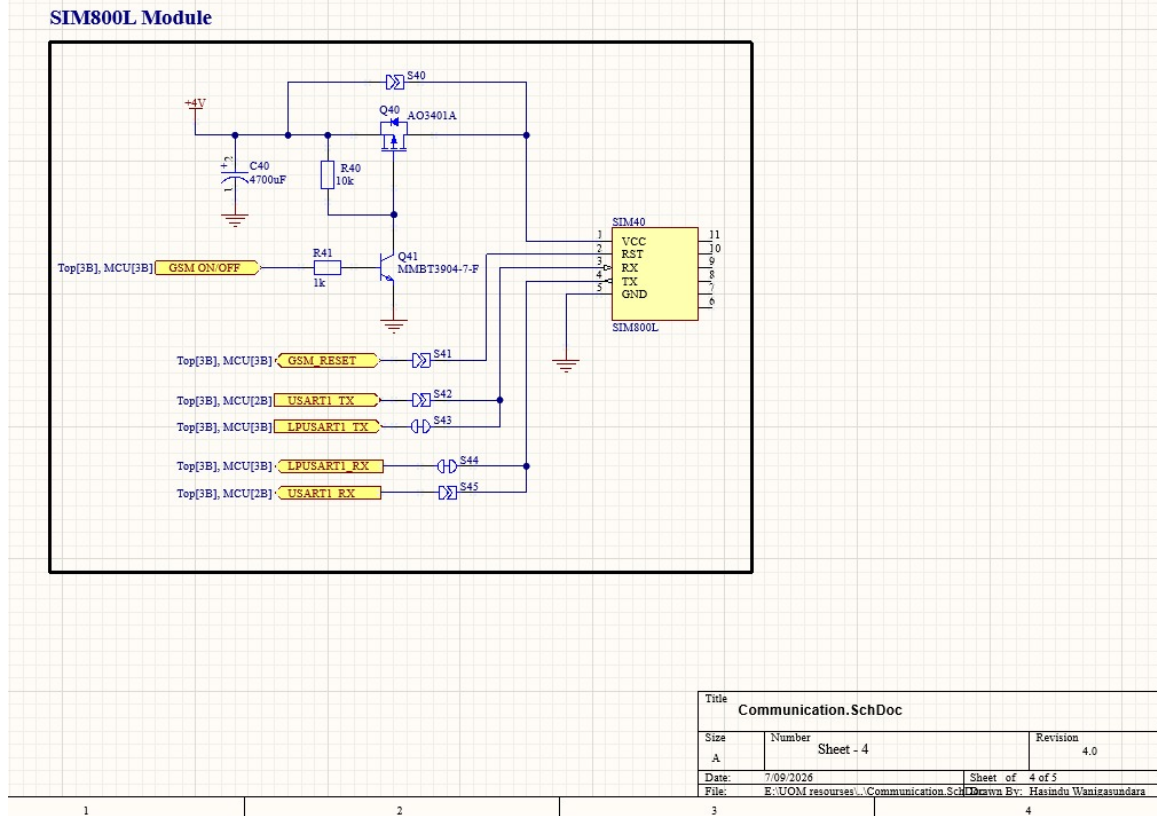


Figure 11: Communication subsystem showing the SIM800L GSM module with switched power control and logic-level protection.

- Hall-effect Sensors:** Three Hall-effect sensor inputs connected to TIM2_CH1–CH3 through B3B-EH JST connectors, each including a 10 k Ω pull-up resistor and local decoupling capacitor.
- OLED Display:** Connected through the I²C1 interface and exposed using both a JST connector and a pin header for displaying system status and water consumption.
- ST-Link Programming Interface:** Dedicated four-pin SWD connector providing firmware programming and debugging capability.
- User Interface:** A green status LED (B50) and a push button provide local diagnostics and manual system interaction.
- Expansion Header:** Additional GPIO pins (PA3, PA4, PB13–PB15) are made available for future hardware extensions such as additional sensors or communication modules.

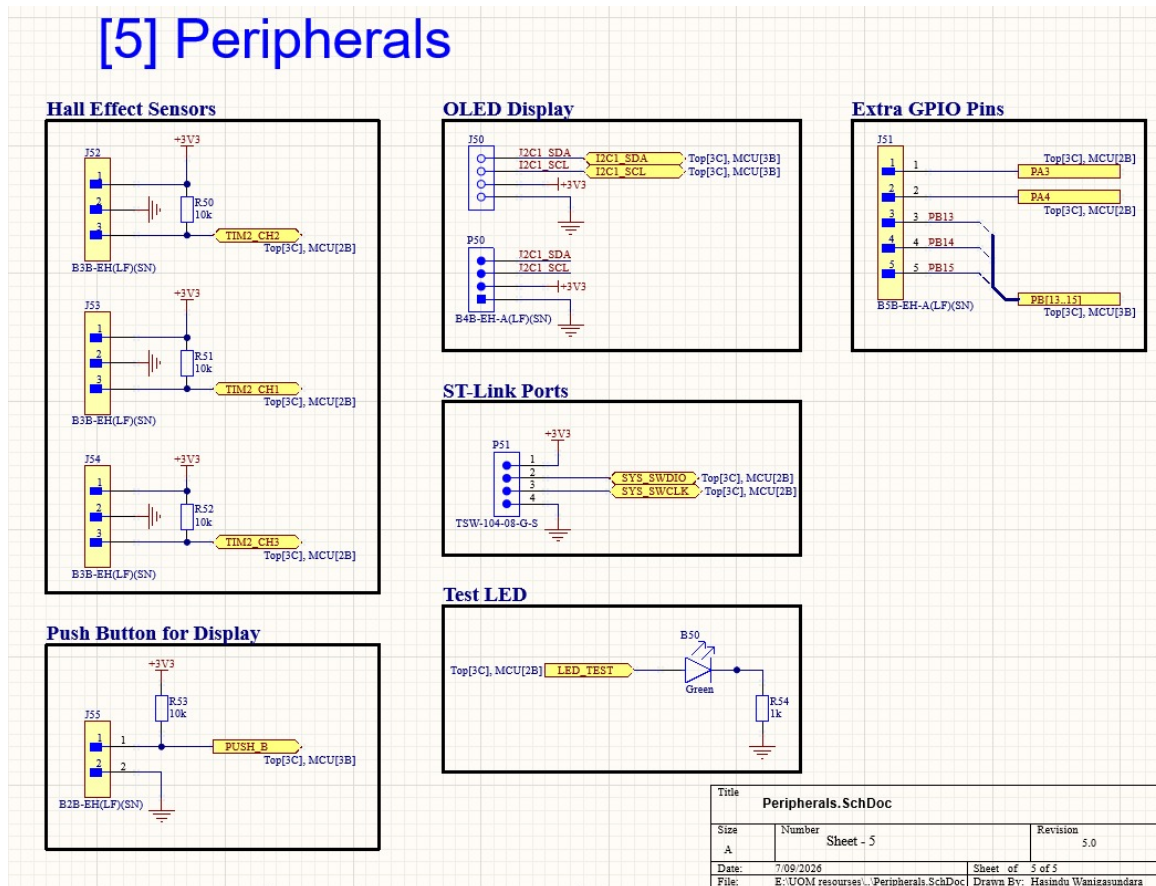
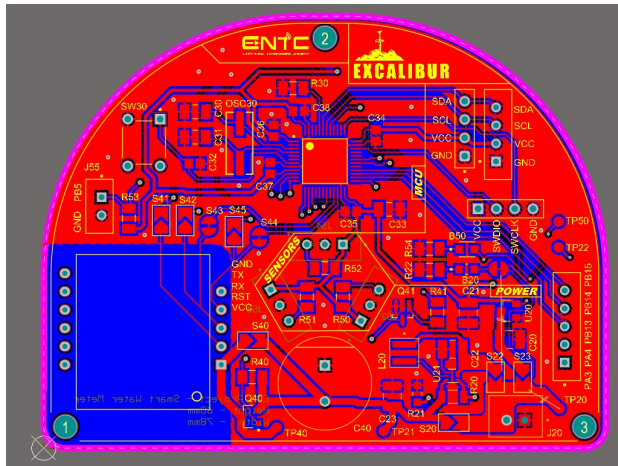
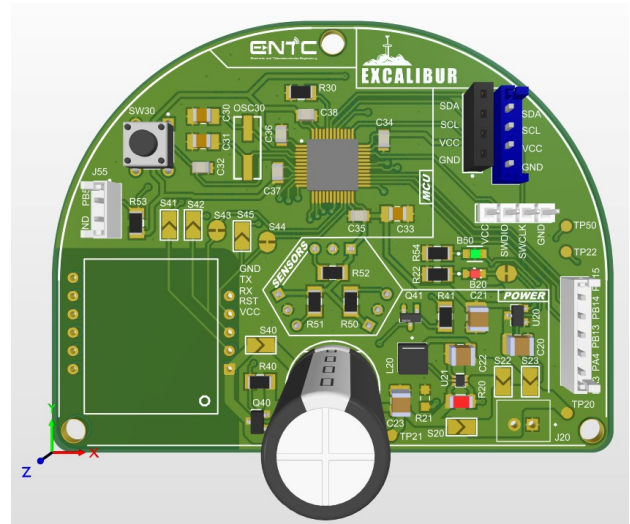


Figure 12: Peripheral subsystem including Hall-effect sensors, OLED display, SWD interface, status LED, push button, and expansion GPIO header.



(a) PCB Layout



(b) 3D View

Figure 13: Final PCB design

7.6 PCB Stack-Up

The designed printed circuit board (PCB) is a standard two-layer FR-4 board selected to achieve a balance between manufacturing cost, mechanical strength, and electrical performance. The

overall board thickness is 1.59 mm, consisting of a 1.50 mm FR-4 dielectric core and 40 μm copper layers on both the top and bottom sides. Solder mask and silkscreen layers are applied to provide electrical insulation, environmental protection, and component identification.

The top copper layer is primarily used for signal routing and component placement, while the bottom copper layer is utilized for additional signal routing and the ground return paths. The FR-4 dielectric provides electrical isolation between the copper layers while maintaining adequate mechanical rigidity for field deployment.

The complete PCB layer stack-up is illustrated in Figure 14.

Table 3: PCB Layer Stack-Up

Layer	Material	Thickness	Purpose
Top Overlay	Silkscreen	–	Component labels
Top Solder Mask	Solder Resist	0.01 mm	Copper protection
Top Copper	Copper	0.04 mm	Signal routing
Core	FR-4 Dielectric	1.50 mm	Electrical insulation
Bottom Copper	Copper	0.04 mm	Signal routing and GND
Bottom Solder Mask	Solder Resist	0.01 mm	Copper protection
Bottom Overlay	Silkscreen	–	Component labels



Figure 14: Layer stack-up of the designed two-layer PCB.

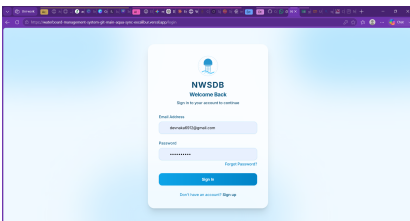
8 Software and Dashboard

To complement the smart water meter hardware, two web-based software platforms were developed: **WaterboardOS** for utility administrators and **AquaSync** for consumers. Together, these platforms provide real-time monitoring, automated billing, data visualization, and centralized system management.

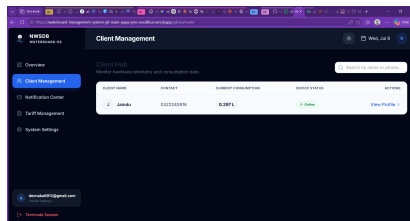
8.1 WaterboardOS (Administrator Dashboard)

WaterboardOS is a centralized web dashboard designed for water utility administrators. The portal can be accessed at: <https://waterboard-management-system-git-main-aqua-sync-excalibur.vercel.app/login>. It provides comprehensive monitoring and management tools for the entire water distribution network.

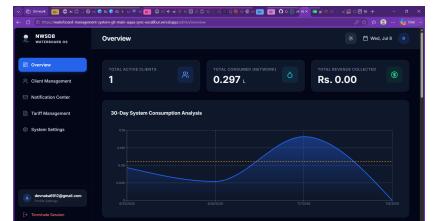
- **System-wide Analytics:** Displays real-time statistics of the total water consumed across the distribution network.
- **Revenue Monitoring:** Provides live tracking of total revenue generated from customer billing.
- **Client Overview:** Displays the total number of active customers connected to the system.
- **Historical Trends:** Visualizes water consumption over the previous 30 days, enabling administrators to identify usage patterns and detect anomalies.
- **Centralized Management:** Provides tools for customer management, notification handling, system configuration, and administrative control.



(a) Login Page



(b) Client Dashboard



(c) Admin Dashboard

Figure 15: Web interfaces developed for the Automated IoT Smart Water Meter system.

8.2 AquaSync (Consumer Web Application)

AquaSync is a consumer-facing web application that enables users to monitor their water usage and billing information in real time. The web application can be securely accessed at: <https://waterboard-management-system-git-main-aqua-sync-excalibur.vercel.app/login>.

- **Personal Usage Tracking:** Displays the total volume of water consumed during the current billing cycle.
- **Live Bill Estimation:** Calculates and updates the estimated water bill continuously based on current consumption.
- **Consumption History:** Presents daily water consumption through interactive graphs for improved usage awareness.
- **Live Synchronization Status:** Indicates the communication status between the smart meter and the cloud, ensuring users receive up-to-date information.
- **Account Management:** Allows users to review previous bills, update profile information, and manage account settings.



Figure 16: Client dashboard interface

9 Mechanical Enclosure Design

The mechanical enclosure was designed to provide a compact, durable, and waterproof housing while ensuring accurate flow measurement and ease of installation within existing municipal water distribution systems. The enclosure integrates the flow sensor, electronics, and plumbing interfaces into a single robust assembly.

- **Standardized Plumbing Integration:** Threaded inlet and outlet ports enable direct, leak-proof connection to existing municipal water pipelines without requiring specialized adapters.
- **Precision Nine-Blade Turbine:**

To optimize the automated smart water meter for high-precision utility billing and long-term durability, the physical rotor was designed with exactly $N = 9$ blades. The engineering justifications for selecting 9 blades over other configurations are summarized below:

- **Elimination of Resonance (Odd vs. Even):** Even-numbered blade configurations (2, 4, 6, 8) cause opposing blades to pass the high-pressure inlet and outlet ports at the exact same instant. This creates symmetric pressure pulses that induce physical vibration, acoustic noise, and premature bearing wear. An odd 9-blade geometry structurally breaks this rotational symmetry, distributing hydrodynamic forces smoothly across the rotation cycle and extending the meter's mechanical lifespan.
- **Maximizing Low-Flow Sensitivity:** Compared to an 8-blade rotor, a 9-blade design increases the physical pulse density by 12.5% per physical revolution. The volumetric flow rate Q is mathematically related to the generated pulse frequency f by:

$$Q = \frac{f}{K} \quad (4)$$

Where K is the meter calibration factor (pulses per unit volume). The pulse frequency f is defined as:

$$f = N \cdot n \quad (5)$$

Where $N = 9$ is the blade count, and n is the rotational speed of the turbine. Increasing N to 9 provides the STM32L0 microcontroller with a higher sampling

resolution, allowing the system to detect minor leaks and low-flow trickles that would otherwise slip through lower blade counts undetected.

- **High Startup Torque:** Lower blade counts (2 to 5) feature wide angular gaps that allow water to bypass the turbine without spinning it. The smaller 40° angular pitch of a 9-blade rotor maximizes hydrodynamic force transfer, allowing the rotor to overcome static bearing friction and begin spinning at minimal flow rates.
- **Prevention of Flow Restriction ($N > 9$):** While increasing the blade count beyond 9 would theoretically increase resolution, it would excessively restrict the physical flow channel, increase hydrodynamic drag (resulting in a higher system pressure drop), and complicate the injection molding process of the physical rotor.
- **Hydrodynamic Cylindrical Chamber:** The internal flow chamber is optimized to minimize turbulence and pressure loss, improving measurement accuracy while maintaining smooth water flow through the meter.
- **Isolated Sensor Mounting:** A waterproof lid secured by four screw bosses isolates the printed circuit board (PCB) and Hall-effect sensors from direct water exposure while maintaining precise alignment with the turbine magnet.
- **Quadrature-Encoding Ready:** The enclosure provides sufficient space for a future dual Hall-effect sensor configuration, enabling quadrature encoding for enhanced flow measurement accuracy and reverse-flow tamper detection.

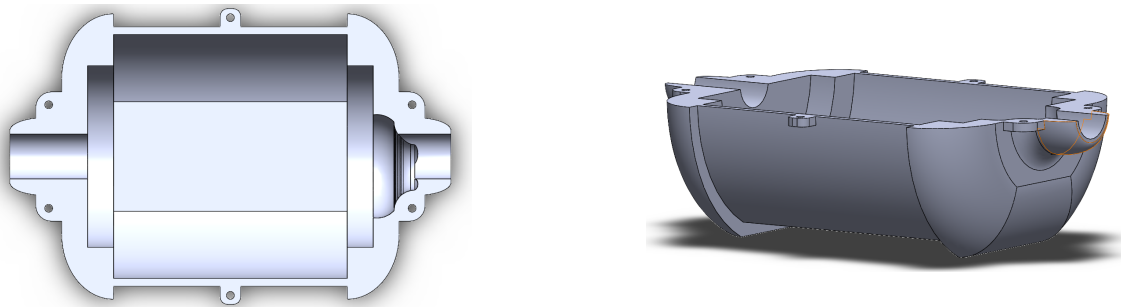


Figure 17: Top and side views of the water meter cover.

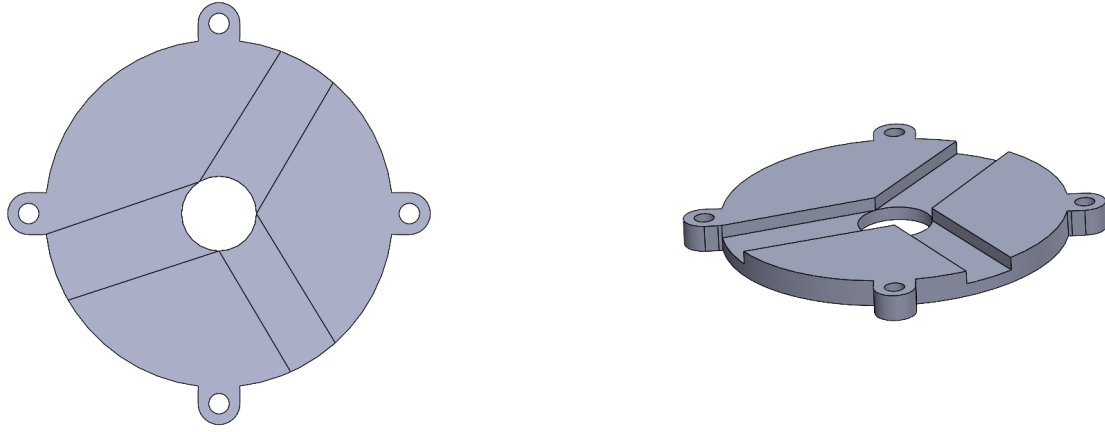


Figure 18: Top and side views of the top lid



Figure 19: Top and side views of the internal assembly

10 Cost Estimation and Production Scope

This section summarizes the estimated prototype cost of the proposed smart water meter and outlines the anticipated improvements and cost reductions achievable through large-scale manufacturing.

Table 4: Estimated Prototype Cost (Single Unit)

Component	Cost (LKR)
STM32L0 microcontroller and supporting components	2,800
SIM800L GSM communication module	1,800
LiSoc12 Er34615	7800
GSM Load Switch AO3401A MOSFET	50
LDO Regulator XC6206	50
Boost Converter TPS61023DRLR	400
PCB fabrication, connectors, and miscellaneous components	5,000
enclosure	3,000
Total Prototype Cost	25,900

10.1 Production Scaling

Large-scale manufacturing significantly reduces the bill of materials (BOM) through bulk component procurement, optimized PCB fabrication, and streamlined assembly processes.

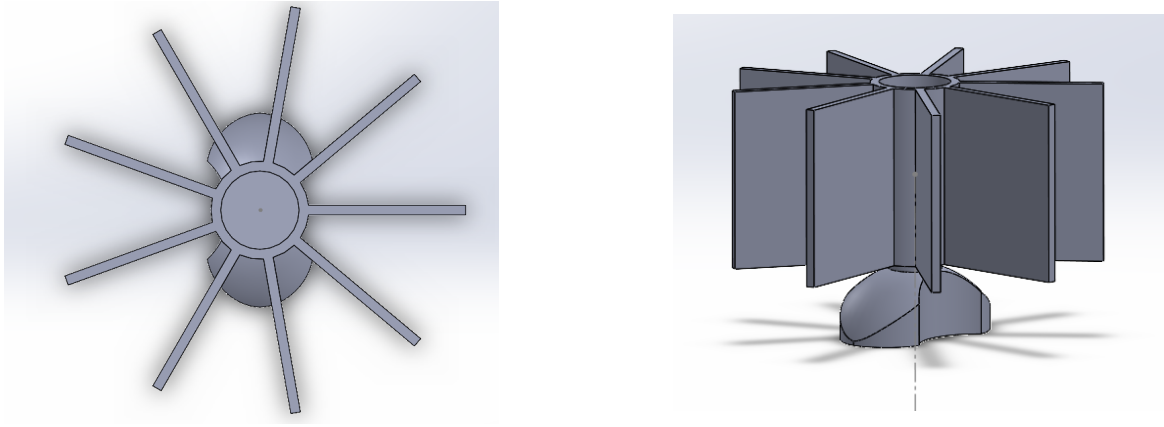


Figure 20: Top and side views of the turbine rotor

- Following table summarizes the estimated total company cost for a production volume of 1 unit.

Table 5: Estimated Total Company Cost per Unit

Cost Item	Cost (LKR)
Manufacturing cost	11,300
Packaging and logistics	700
Cloud hosting and database services (1 year)	500
Warranty and technical support	1,000
Administrative and operational expenses	1,500
Total Company Cost	15,000

- Bulk purchasing of electronic components to reduce overall BOM cost.
- Custom-designed PCB replacing development modules.
- Industrial-grade IP67 weatherproof enclosure for outdoor deployment.
- Migration from GSM-only communication to LoRaWAN or NB-IoT for improved scalability and lower communication costs.
- Production cloud infrastructure with 99.9% service availability, secure JWT-based authentication, and enterprise-grade data management.

The transition from prototype to commercial deployment will require collaboration with the NWSDB, including pilot installations, product validation, Memoranda of Understanding (MoUs), and formal type-approval certification.

11 Team Contributions

The successful design and implementation of the Automated IoT Smart Water Meter were made possible through the collaborative efforts of all team members. The individual responsibilities and technical contributions of each member are detailed in Table 6.

Table 6: Team Member Contributions

Index No.	Name	Contributions
230680M	W.M.H. Wanigasundara	PCB design (MCU, Sensor), Soldering, PCB Testing, Debugging
230689A	A.H.T.M. Weerakoon	Solidworks Design, Soldering, Integration and Testing, Component selection
230495B	H.W.D. Prabarshana	Solidworks Design, Documentation, Procurement
230395T	K.D. Manatunga	Solidworks Design, Soldering, PCB Testing, Integration
230563H	H.D.J.D. Samaranayaka	PCB design (power), Documentation, Web application, Web testing

12 Conclusion

The proposed Automated IoT Smart Water Meter offers a practical and cost-effective solution to Sri Lanka's Non-Revenue Water (NRW) challenge by replacing conventional manual metering with an intelligent, IoT-enabled monitoring and billing system. By integrating low-power embedded hardware, GSM communication, cloud-based data management, and user-friendly web dashboards, the system enables real-time water consumption monitoring, automated billing, and early leak detection while significantly reducing operational costs and improving billing accuracy.

The modular hardware architecture, long-life battery-powered operation, and scalable cloud infrastructure make the system suitable for both residential and utility-scale deployment. Furthermore, the mechanical enclosure has been designed to ensure reliable operation in outdoor environments while supporting future hardware expansion.

With the PCB design, embedded firmware, cloud backend, and mechanical enclosure already under development, Team Excalibur is well positioned to deliver a fully functional and validated prototype by the conclusion of the EN2160 Engineering Design Realization module. The proposed design also provides a clear pathway toward large-scale manufacturing and deployment, contributing to improved water resource management, reduced Non-Revenue Water losses, and the digital transformation of Sri Lanka's water distribution infrastructure.

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